

Abhishek Reddy Manam

Data Engineer · Azure Data Factory · Databricks · PySpark · Delta Lake · Snowflake · dbt · Kafka · Airflow

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PROFESSIONAL SUMMARY

Azure-focused Data Engineer with 4+ years of experience designing and delivering enterprise-scale data platforms across Azure, AWS, and GCP. Deep hands-on expertise in Azure Data Factory, Azure Databricks, PySpark, Delta Lake Medallion Architecture (Bronze/Silver/Gold), Snowflake, dbt, Apache Kafka, and Apache Airflow. Proven track record of reducing pipeline processing latency by 50%, cutting data quality incidents by 40%, shrinking data-source onboarding from days to hours through metadata-driven pipeline design, and building real-time streaming systems processing 5M+ events/day with exactly-once semantics. Passionate about lakehouse architecture, analytics engineering, DataOps, and translating raw multi-source enterprise data into reliable, governance-ready, business-ready insights.

TECHNICAL SKILLS

Azure: Azure Data Factory (ADF) — metadata-driven pipelines, parameterised templates, linked services, Mapping Data Flows, IR; Azure Databricks — PySpark, Delta Live Tables, Unity Catalog, cluster management; Azure Synapse Analytics — Serverless SQL Pool, OPENROWSET, External Tables; ADLS Gen2; Azure Event Hubs; Azure DevOps CI/CD; Azure Purview; Azure Monitor; Microsoft Fabric

AWS: S3 · Glue (ETL jobs, crawlers, data catalog) · Redshift · EMR (managed Spark) · Lambda · Kinesis · Step Functions · IAM · CloudWatch

GCP: BigQuery · Dataflow (Apache Beam) · Pub/Sub · Cloud Composer (managed Airflow) · Cloud Storage

Data & ETL: Snowflake (virtual warehouses, Snowpipe, Streams & Tasks, RBAC, masking policies) · dbt (staging/intermediate/mart layers, schema tests, SCD Type 2 snapshots, incremental materialisation, slim-CI) · Apache Kafka (exactly-once semantics, windowed aggregations, dead-letter queues) · Apache Airflow (dynamic DAG generation, SLA monitoring, PagerDuty/Slack alerting) · PySpark · Delta Lake (Z-ORDER, ACID, time-travel) · Great Expectations · Star Schema · Snowflake Schema · Data Vault

Languages: Python (Advanced) — ETL scripting, Pandas, NumPy, boto3, pytest; SQL (Advanced) — complex queries, CTEs, window functions, query optimisation; Bash · YAML · JSON

DevOps & IaC: Docker · Terraform (Azure & AWS resource provisioning) · GitHub Actions · Azure DevOps CI/CD · Git · Kubernetes (familiar)

Architecture: Medallion Architecture (Bronze/Silver/Gold) · Lambda Architecture · Lakehouse Design · Metadata-Driven Pipeline Frameworks · Event-Driven Architecture · Data Mesh (familiar) · OLAP/OLTP design trade-offs

PROFESSIONAL EXPERIENCE

Data Engineer • *Digiuniv Technologies · Hyderabad, India*

Oct 2021 – Jul 2024

- Architected a fully metadata-driven Azure Data Factory framework — a single parameterised pipeline template reads source/target configs from a SQL control table, replacing 200+ hardcoded pipelines across 40+ linked services (Azure SQL, ADLS Gen2, Salesforce, REST APIs, SAP); cut new data-source onboarding from 3 days to under 2 hours and reduced pipeline deployment effort by 85%.
- Engineered high-performance PySpark transformation workloads on Azure Databricks — applied broadcast joins, predicate pushdown, partition pruning, and Adaptive Query Execution (AQE) on a Delta Lake Medallion Architecture (Bronze → Silver → Gold) on ADLS Gen2, achieving 50% reduction in end-to-end pipeline latency and enabling sub-hourly SLA compliance for 15 business-critical KPIs.
- Owned the complete dbt transformation layer in Snowflake — 80+ staging, intermediate, and mart models with ref() lineage, schema tests (not_null, unique, relationships, accepted_values), SCD Type 2 snapshots for 6 dimension tables, and incremental

materialisation for large fact tables; CI/CD-gated deployment via GitHub Actions slim-CI reduced data quality incidents by 40% and full-refresh runtimes by 60%.

- Built the organisation's first Apache Kafka + PySpark Structured Streaming pipeline processing 5M+ events/day with exactly-once semantics; consumed from Azure Event Hubs Kafka-compatible endpoint, applied windowed aggregations and sessionisation, sinked to Azure Synapse Analytics achieving end-to-end latency under 90 seconds.
- Migrated batch Airflow to dynamic DAG generation on GCP Cloud Composer spanning BigQuery, Dataflow, and AWS Glue — implemented SLA miss callbacks, PagerDuty + Slack alerting, and idempotent retry-with-backoff logic cutting MTTR from 2 hours to 30 minutes.
- Containerised all pipeline services with Docker, provisioned Azure infrastructure via Terraform IaC, and built full Azure DevOps CI/CD pipelines with automated unit tests, schema linting, and blue/green deployment gates for zero-downtime releases.
- Established Great Expectations validation suites as mandatory pre-Gold-layer build gates — automatically catching schema drift, null anomalies, and referential integrity failures before data reached downstream Power BI dashboards consumed by 200+ business users.
- Mentored 3 junior engineers via architecture walkthroughs, code reviews, and pair-programming; authored internal runbooks on Medallion Architecture, dbt best practices, and ADF metadata-driven design patterns adopted across the team.

Associate Data Engineer • *Digiuniv Technologies · Hyderabad, India*

Jun 2020 – Oct 2021

- Took full ownership of end-to-end data pipelines — built dynamic ADF workflows with JSON-driven configuration scaling across heterogeneous sources without per-integration code changes, accelerating delivery velocity and reducing pipeline sprawl.
- Implemented Medallion Architecture (Bronze, Silver, Gold) with Delta Lake on ADLS Gen2 using PySpark in Azure Databricks — type casting, null handling, derived columns, surrogate key generation, and multi-source joins for a retail analytics domain.
- Served business-ready Gold-layer data through Azure Synapse Serverless SQL Pool via OPENROWSET, SQL Views, and External Tables with Managed Identity, feeding Power BI dashboards used by 200+ stakeholders across finance, operations, and executive teams.
- Led dbt Snowflake project from scratch — staging, intermediate, and mart layers with incremental materialisation and schema tests; reduced full-refresh runtimes by ~60%; established team coding conventions for model naming and testing.
- Designed and deployed Apache Kafka + PySpark Structured Streaming pipelines for high-volume clickstream data with exactly-once delivery semantics, windowed aggregations, and dead-letter queue handling.
- Owned Airflow DAG development — dynamic generation, SLA monitoring, and Slack alerting; containerised all services with Docker and integrated CI/CD pipelines for automated testing and deployment.

Data Engineering Intern • *Digiuniv Technologies · Hyderabad, India*

Apr 2020 – Jun 2020

- Contributed to production-scale ETL pipelines on AWS (S3, Glue, Lambda, Step Functions) and GCP (BigQuery, Pub/Sub, Cloud Storage) — hands-on exposure to multi-cloud ingestion patterns alongside senior engineers.
- Developed and maintained Apache Airflow DAGs for multi-step workflows spanning AWS Glue jobs and BigQuery loads with dependency management and Slack failure alerting.
- Wrote PySpark transformation scripts on AWS EMR for large-scale deduplication, null-handling, and schema enforcement on raw ingestion datasets.
- Built Python utilities for automated row-count reconciliation and data quality validation between source systems and target Snowflake tables.

KEY PROJECTS

Enterprise Metadata-Driven Lakehouse Platform | *Azure Data Factory · Databricks · Delta Lake · Synapse Analytics · Power BI · Terraform*

- Designed a fully metadata-driven ADF orchestration framework — a single parameterised pipeline template reads source/target configs from a SQL control table, reducing pipeline deployment effort by 85% and eliminating pipeline sprawl across 40+ linked services (Azure SQL, Blob, Salesforce, REST APIs, SAP).

- Built a 3-tier Delta Lake Medallion Architecture on ADLS Gen2 with schema evolution, Z-ORDER clustering, Liquid Clustering, and VACUUM strategies; delivered sub-hourly data freshness for 15 mission-critical KPIs to C-suite and operations teams via Azure Synapse + Power BI DirectQuery — reducing report refresh time from 4 hours to under 30 minutes.

Real-Time Event Streaming Platform | *Apache Kafka · PySpark Structured Streaming · Azure Event Hubs · Azure Synapse · Dead-Letter Queue*

- Architected an end-to-end Kafka pipeline ingesting 5M+ events/day from Azure Event Hubs with exactly-once semantics, fault-tolerant checkpointing, and automatic offset management across 12 consumer groups.
- Applied PySpark windowed aggregations (tumbling + sliding windows) for real-time sessionisation, fraud signal detection, and live inventory updates with sub-90-second end-to-end latency; built dead-letter queue handling, schema registry validation, and automated Slack alerting for consumer lag SLA breaches.

Analytics Engineering Layer — dbt + Snowflake | *dbt · Snowflake · GitHub Actions CI/CD · Great Expectations · Python*

- Built 80+ dbt models across staging, intermediate, and mart layers with full ref() dependency DAG, schema tests, source freshness checks, and SCD Type 2 snapshots for 6 slowly-changing dimension tables; incremental materialisation across all large fact tables cut full-refresh runtimes by 60%.
- Deployed Great Expectations as a mandatory post-dbt quality gate; automated anomaly detection prevented 3 major bad-data incidents from reaching downstream reporting in 6 months. Slim-CI in GitHub Actions (--select state:modified+) reduced CI build time by 70%.

Cross-Cloud Airflow Orchestration — GCP Cloud Composer | *Apache Airflow · GCP Cloud Composer · BigQuery · AWS Glue · Dataflow · PagerDuty*

- Migrated static DAGs to a dynamic DAG factory pattern reading configs from GCS JSON files — reduced DAG file count by 70% while expanding orchestrated pipeline coverage; implemented SLA callbacks, PagerDuty + Slack alerting, and retry-with-backoff logic cutting MTTR from 2 hours to 30 minutes.
- Designed idempotent cross-cloud task graphs spanning GCP BigQuery, Dataflow, and AWS Glue crawlers with dependency-aware scheduling and full audit logging for compliance and reproducibility.

EDUCATION

M.S. Information Systems Technologies · Wilmington University, Newark, DE · GPA 3.6 / 4.0 *Aug 2024 – May 2026*

B.Tech. Computer Science & Engineering · Lovely Professional University, Punjab, India · GPA 7.2 / 10 *Graduated 2020*